

AI –POWERED CHATBOT FOR CUSTOMER SUPPORT

1. Project overview:

The project aims to build a AI-powered Chatbot that helps to reduce the workload of customer service team and to reduce the response time of customer enquiries.

2. Business problem :

A-Ecommerce company facing a high volume of repetitive customer enquiries for orders, return and refund handling these is work manually is time consuming and exhausting that leads to increasing in customer-service workload, operational costs and response times.

3. Business objective:

The objective is to reduce customer-service workload and improve the handling of customer enquiries.

- Reduce the volume of repetitive customer-service enquiries handled manually by 5% within one month of chatbot launch.
- Ensure that at least 80% of supported customer enquiries receive a response within the defined response-time target within one month of chatbot launch.

4. Stakeholders:

The stakeholders involved in the project are

Stakeholder	Role	power	Interest
Customer	Use the chatbot to clarify their enquiries.	Low	Low
Customer service team	They handle the enquiries of the customers.	Low	High
Product owner	They own the product requirement and priorities.	High	High

Developing team	They build the chatbot functionalities.	Medium	High
QA team	They test the functionality of the chatbot.	Medium	High
Devops team	They deploy the chatbot.	High	High
AI/ML team	They check and train the AI/ML functionalities in chatbot.	High	High
Product manager	They improve and give new updates on the feature.	High	High
Business analyst	They handle all the requirement of the features	High	High

5. Scope:

5.1. In scope:

- FAQ handling.
- Order enquiries.
- Return and refund related enquiries.
- Customer authentication before accessing account/order information
- Access to authorized order information
- Human escalation for unsupported/complex queries
- Basic chatbot performance monitoring

5.2. Out scope:

- Customer enquiries using their regional language.
- Implement a voice based enquiries for customer.
- Autonomous high-risk decisions/refunds.

6. Business requirement:

BR01:The business needs to reduce and handle repetitive customer enquiries.

BR02:The business needs to improve the response time of customer enquiries.

BR03:The business need to provide accurate answer for predefined FAQ .

BR04:The business needs to escalate to human customer service agent when the question is too complex to handle.

BR05: The business needs customer account information to be accessed only after appropriate customer authentication and authorization, while protecting customer privacy and security.

BR06: The business needs customers to have self-service access to order, return, and refund information at any time.

BR07: The business needs to measure chatbot performance and customer-support outcomes to evaluate the effectiveness of automating customer support.

BR08: The business needs to use chatbot performance reports to identify areas for improvement and support continuous improvement of customer service.

7. Functional Requirements:

FR01 – The chatbot shall provide answer when customer has enquiry on their order,refund and return details.

FR02 – The chatbot shall provide responses to supported customer enquiries within the defined response-time requirement.

FR03 – The chatbot shall answer most frequently asked question asked by customer for enquiry.

FR04 – When the chatbot can't answer a complex customer enquiry then it shall escalate it to human customer service agent.

FR05 – The chatbot shall verify customer authentication before providing any account-specific information.

FR06 - The chatbot shall allow authenticated customers to request and view their authorized order, return, and refund information.

FR07 – The system shall provide an administrative dashboard that allows authorized administrators to monitor chatbot performance and customer-support outcomes.

8. Non Functional Requirements:

NFR01 : The system shall provide response to customer enquiry within 2-5 seconds under normal operating conditions.

NFR02 : The system shall collect and process only minimum customer account information required to order, return, and refund.

NFR03 : The chatbot shall be available **24/7**, subject to planned maintenance and service interruptions.

NFR04: The system interface shall be simple and intuitive so that it can be easy to submit their enquiries and understand response.

9. User story:

US01 : As a customer I want to know ask chatbot about order details so that I can know when my order arrives.

US02 : As a customer I want to ask chatbot about return of a product so that I can know about product return window.

US03 :As a customer I want to ask chatbot about refund so that I can know when to expect my refund.

US04 :As a customer I want to contact human customer service agent when the chatbot cannot resolve my enquiry so that I can receive further assistance.

US05 :As a customer I want to ask FAQ to chatbot so that I can get quick answers without contacting customer service agent.

US06 : As a customer I want my conversation context available to the human customer service agent after escalation so that I do not have to repeat my enquiry.

US07 : As a customer, I want to access my authorized account information through the chatbot, so that I can ask about my orders, returns, and refunds.

US08 :As a Administrator I want to monitor the chatbot so that I can improve the chatbot response.

10.Acceptance criteria :

AC01.1 : Given the customer is authenticated and has a valid order, **when** the customer asks the chatbot for their order details, **then** the chatbot shall display the relevant order status and estimated delivery information.

AC01.2 : Given the customer is authenticated provides a invalid order or non existent order-id, **when** the customer asks the chatbot for their order details, **then** the chatbot shall inform the customer that the order could not be found.

AC02.1 : Given the customer is authenticated has a valid order **when** the customer asks the chatbot for their product return window details, **then** the chatbot shall display the applicable return window and return details.

AC02.2 : Given the customer is authenticated has a invalid or non existent order id **when** the customer asks the chatbot for their product return window details, **then** the chatbot shall inform the customer that the order could not be found.

AC03.1 : Given the customer is authenticated has a valid refund request **when** the customer asks the chatbot for their product refund details, **then** the chatbot shall display the relevant order details ,return details and the expected refund information.

AC03.2 : Given the customer is authenticated has a invalid or unavailable refund request reference **when** the customer asks the chatbot for their product refund details, **then** the chatbot shall inform the customer that the refund request could not be found .

AC04.1 : Given the customer has a enquiry that is supported by the chatbot **when** the customer asks the chatbot the enquiry, **then** the chatbot shall display the relevant response.

AC04.2 : Given the customer has an enquiry **when** the chatbot cannot provide reliable for customer enquiry, **then** the chatbot shall inform it cannot provide a reliable answer and provide an option to escalate it to the human customer service agent.

AC05.1 : Given the customer asks a supported FAQ **when** the customer asks the chatbot for that enquiry, **then** the chatbot shall provide the relevant answer based on the available FAQ information.

AC05.2 : Given the customer asks a unsupported FAQ ,**when** the chatbot cannot provide the answer ,**then** the chatbot shall provide an option to escalate it to human customer service agent.

AC06.1 : Given the customer has an enquiry and interacted with the chatbot **when** the chatbot cannot provide the answer for enquiry and escalates it, **then** the conversation context shall be escalated to human customer service agent.

AC06.2 : Given the customer told their order id **when** customer contacts the human customer service agent , **then** the conversation context including order id shall be available to human customer service agent.

AC07.1 : Given the customer is authenticated **when** the customer want to view account specific order, return and refund details, **then** the chatbot shall display only the information authorized for that customer.

AC07.2 : Given the customer is not authenticated **when** the customer want to view account specific order, return and refund details, **then** the chatbot shall not display the requested information and prompt the customer to authenticate.

AC08.1 : Given the administrator is monitoring chatbot performance, **when** the chatbot provides an incorrect response, **then** the system shall allow the administrator to record and report the incorrect response for review.

AC08.2 : Given the administrator is monitoring chatbot performance, **when** the administrator identifies a response that requires improvement, **then** the system shall allow the administrator to record the issue for analysis and improvement.

11.AI / Data Requirement :

11.1. Data requirement :

DR01 :The system shall require a customer order data to help customer with their order details, order status and estimated delivery information.

DR02 : The system shall require a customer refund data to provide refund details, expected refund information.

DR03 : The chatbot shall require FAQ data to provide answers to common customer enquiries.

DR04 :The chatbot shall require authorized customer account data to provide order,return and refund information.

DR05 : The chatbot shall require conversation data to maintain conversation context and provide relevant context to the customer service agent after escalation.

DR06 : The system shall require customer return data to provide return status, return eligibility, and applicable return-window information.

11.2. Data quality requirement :

DRQ01: order data shall be complete and accurate so that the chatbot can provide order details, and estimated delivery information .

DRQ02: Refund data shall be complete and accurate so that the chatbot can provide the customer with refund status and expected refund information .

DRQ03: The FAQ data shall be complete and accurate to provide the customer with correct information.

DRQ04: Authorized customer account data shall be accurate to provide the customer with correct order, return, and refund information.

DRQ05: Conversation data shall be complete and accurate so that the relevant conversation context can be provided to the human customer-service agent after escalation.

DRQ06: Return data shall be complete and accurate so that the chatbot can provide the customer return status and return window information.

11.3. AI Performance Requirements :

AI01:The chatbot shall achieve at least 90% accuracy when answering predefined FAQ questions

AI02:The chatbot shall provide at least 80% of accurate responses on authorized customer account specific information.

AI03:The chatbot shall provide at least 80% accurate responses to customer enquiries related to order, return, and refund information.

AI04:The chatbot shall be able to answer 85% of the supported customer enquiry completely and accurately.

AI05:The chatbot shall correctly retain relevant conversation context during a customer session and make the relevant context available to the human customer-service agent after escalation.

11.4. AI Fallback & Human Escalation Requirements:

AF01: The chatbot shall provide a fallback response when it cannot reliably answer a customer enquiry.

AF02: The chatbot shall not generate or provide an answer based on unsupported or insufficient information.

AF03: The chatbot shall not make autonomous high-risk decisions, such as approving or rejecting a customer refund.

AF04: When the chatbot cannot reliably answer an enquiry, it shall provide an option to escalate the enquiry to a human customer-service agent.

AF05: When an enquiry is escalated, the chatbot shall transfer the relevant conversation context to the human customer-service agent.

11.5. AI Security, Privacy & Responsible AI Requirements:

SR01: The chatbot shall protect customer personal information from unauthorized access.

SR02: The chatbot shall collect and process only the customer account information that are necessary for handling enquiries.

SR03: The chatbot shall provide account specific information only to the authorized customer account.

SR04: The chatbot shall inform the customer that they are interacting with AI-chatbot before the conversation begins.

SR05: The system shall retain relevant conversation with chatbot to monitoring, accountability and reviewing in accordance with applicable data-retention requirements.

12. BPMN/Process flow :

12.1. Process Overview :

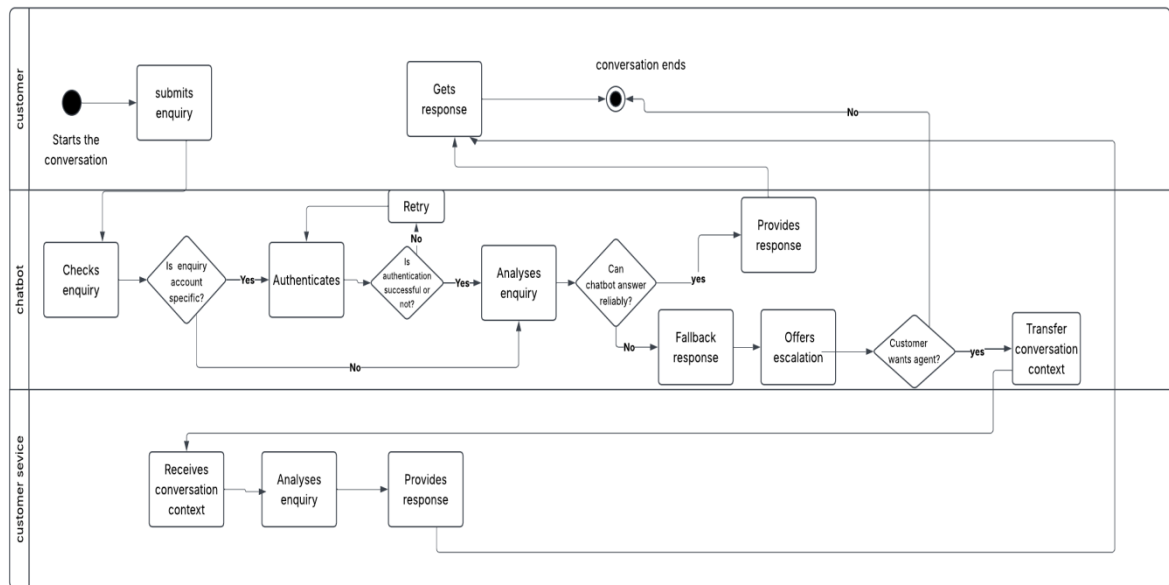
The following BPMN process flow illustrates how a customer enquiry is handled by the AI chatbot, including account authentication, chatbot response, fallback, and escalation to a human customer-service agent.

12.2. Process Description

Step	Process
1	Customer submits an enquiry
2	Chatbot analyzes the enquiry
3	Account-specific enquiries require authentication
4	Chatbot determines whether it can reliably answer
5	If it can answer, the chatbot provides a response
6	If it cannot reliably answer, fallback is provided

7	Customer can request human-agent escalation
8	Relevant conversation context is transferred to the human agent
9	Human agent handles the enquiry and responds

12.3. BPMN process flow :



13. AI Risk & Ethics :

Risk Id	Risks	Impact	Mitigation
R01	The chatbot may provide different quality or accuracy of responses to different customer groups because of biased training data.	This could cause some customer groups to receive poorer-quality or less accurate responses.	The chatbot shall be trained and tested using diverse customer data, and its performance shall be monitored across different customer groups.
R02	The chatbot may take autonomous high risk	This could leads to financial loss.	The chatbot shall not take autonomous high risk decision on its own and it

	decision of approving the refund for non existing order.		needs to escalate it to human customer service agent.
R03	Customer may not know they are making conversation with AI	Customers may misunderstand the source or reliability of the response and may not know when human assistance is available.	The chatbot shall inform the customer that they are interacting with AI chatbot and provide an option for human escalation.
R04	The chatbot may provide response to customer when it is not sure of response/with low knowledge.	Customer may get unreliable answer and get misleading information.	The chatbot shall not provide response to customer when it is not sure of response/with low knowledge and provide option to escalate it to human customer service agent.
R05	The chatbot may expose customer account information to unauthorized users.	This could result in privacy violations and loss of customer trust.	The chatbot shall authenticate and authorize customers before providing account-specific information and shall prevent unauthorized access to customer data.

14. KPIs / Success Metrics :

KPI ID	Success metric	How it measured?	Target
KP01	FAQ accuracy	Out of 100 FAQ the chatbot shall answer 90 or above.	The target of FAQ is $\geq 90\%$
KP02	Customer satisfaction	Percentage of positive responses from post-chat customer satisfaction surveys	The target shall be $\geq 85\%$
KP03	Account-specific information	Measured by the accuracy of the specific account	The target shall be $\geq 80\%$.

		information.	
KP04	Response Time	The Average response time taken by the chatbot.	The target shall be 2-5 seconds.
KP05	Human escalation rate	Escalated enquiries ÷ total chatbot enquiries × 100	Baseline to be established during launch.
KP06	Conversation context retention	Escalated conversations where relevant context is correctly transferred to the human agent ÷ total escalated conversations × 100	The target Shall be >=70%.

15. Test cases :

Test Case ID	Requirement / User Story	Test Scenario	Test Steps	Expected Result	Status
TC01	US01	Valid order id	Click chatbot, authenticate, enter valid order id	The chatbot shall display order details	Not Run
TC02	US01	Invalid order id	Click chatbot, authenticate, enter invalid order id	The chatbot shall display order could not be found	Not Run
TC03	US02	Valid product return window	Click chatbot, authenticate, enter return details with valid order id	The chatbot shall display	Not Run

				return details	
TC04	US02	Invalid product return window	Click chatbot,authenticate,enter return details with invalid order id	The chatbot shall inform the customer that the order could not be found	Not Run
TC05	US03	Valid product refund	Click chatbot, authenticate, and enter a valid refund request reference	The chatbot shall display the relevant order details, return details, and expected refund information.	Not Run
TC06	US03	Invalid product refund	Click chatbot, authenticate, and enter an invalid or unavailable refund request reference	The chatbot shall inform the customer that the refund request could not be found.	Not Run
TC07	US05	Supported FAQ	Click chatbot,enter supported enquiry	The chatbot shall provide a relevant answer based on the available	Not Run

				FAQ information.	
TC08	US05	Unsupported FAQ	Click chatbot ,enter unsupported FAQ	The chatbot shall display can't know the answer and provide human escalation	Not Run
TC09	US06	Conversation escalation	Contact customer support,tell order id	The conversation context is given for customer service agent.	Not Run
TC10	US07	Authorized access	Click chatbot,authenticate,ask order or refund details	The chatbot shall display account specific order or refund details	Not Run
TC11	US07	Unauthorized access	Click chatbot,ask order or refund details	The chatbot shall display have to authenticate to access account details	Not Run
TC12	US08	Monitor performance	Identifies a response that needs improvement	The system shall allow the administrator to	Not Run

				record the issue for analysis and improvement	
TC13	NFR01	Chatbot response time	Submit a supported customer enquiry and record the time taken for the chatbot to respond.	The chatbot shall provide a response within 2–5 seconds under normal operating conditions.	Not Run

16. Power BI Dashboard :

16.1. Dashboard Objective :

The objective of the Power BI dashboard is to monitor the AI chatbot's performance, customer satisfaction, response time, human escalation, and conversation context retention. The dashboard will help administrators and business stakeholders identify performance issues and areas for improvement.

16.2. Dashboard Dataset :

The dashboard will use sample chatbot conversation data to monitor chatbot performance and calculate the defined KPIs. The dataset will contain conversation details, enquiry type, response accuracy, customer satisfaction, response time, escalation status, account-information accuracy, and conversation-context transfer.

16.3. Insights :

- The chatbot achieved 100% FAQ accuracy, exceeding the target of 90%. This indicates strong performance when handling predefined FAQ enquiries.

- The chatbot achieved 70% customer satisfaction, which is below the target of 85%. This indicates that the customer experience requires improvement.
- The chatbot achieved 62.5% account-specific information accuracy, which is below the target of 80%. This indicates that account-specific responses require improvement in accuracy and reliability.
- The chatbot's average response time was 4 seconds, which meets the defined target of 2–5 seconds. This indicates that the chatbot is meeting the expected response-time requirement.
- The human escalation rate was 16.7%. This can be used as the initial baseline for monitoring future chatbot performance and identifying frequently escalated enquiry types.
- Conversation context retention was 80%, exceeding the target of 70%. This indicates that relevant conversation context is successfully transferred to human customer-service agents in most escalated conversations.

16.4. Recommendations :

- Improve customer satisfaction by reviewing conversations with low satisfaction ratings, identifying recurring customer issues, and improving chatbot responses to prevent similar issues from repeating.
- Improve account-specific information accuracy by reviewing incorrect responses and validating the integration between the chatbot and customer account data sources.
- Maintain FAQ performance by continuously monitoring FAQ accuracy and regularly updating the FAQ knowledge base with relevant customer enquiries and new frequently asked questions.
- Monitor human escalation by using the 16.7% escalation rate as a baseline and analyzing frequently escalated enquiry types to identify opportunities for chatbot improvement.
- Maintain conversation-context retention by continuously monitoring context transfers and investigating failed transfers to maintain context retention above the 70% target.

16.5. Dashboard Conclusion :

The Power BI dashboard provides stakeholders with visibility into chatbot performance, customer satisfaction, response time, human escalation, and conversation context retention. The sample results indicate strong FAQ accuracy, acceptable response time, and effective conversation context transfer. However, customer satisfaction and account-specific information accuracy are below their defined targets and should be prioritized for improvement.

